The Art of Uncertain Topoplots: Designing and Evaluating Uncertainty Visualizations for EEG/ERP Data

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Abstract—TODO: Supplementary-material overview

Index terms—TODO: Supplementary Material

1 Supplementary Material

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Manuscript received DD MMM. YYYY; accepted DD MMM. YYYY. Date of Publication DD MMM. YYYY; date of current version DD MMM. YYYY. For information on obtaining reprints of this article, please send e-mail to: reprints@ieee.org. Digital Object Identifier: xx.xxxx/TVCG.YYYY.xxxxxxx

ID	Interview question
Warm-up	
Q1	Could you briefly describe your background in EEG/ERP research?
Q2	What types of EEG visualizations do you most commonly use in your work, and for what purposes?
Thinking About Uncertainty	
Q3	When you think about quantifying uncertainty in EEG or ERP data, what kinds of methods come to mind?
Q4	Do you currently include uncertainty information when you visualize EEG/ERP data? If so, what type of uncertainty do you include, and why?
Use of Topoplots	
Q5	At what stages of your research do you use topoplots, and for which audiences?
Q6	When you create a topoplot, what purposes does it serve?
Q7	Why do you think other researchers use topoplots?
Uncertainty in Topoplots	
Q8	Do you ever show uncertainty in topoplots specifically? If not, why not?
Q9	If you wanted to show uncertainty in a topoplot, how might you visualize it?
Value and Barriers	
Q10	Do you think visualizing uncertainty in topoplots would be helpful? Why or why not?
Q11	Why do you think many researchers currently do not include uncertainty in their topoplots?
Future Needs	
Q12	Is there anything you wish topoplot tools did better?
Q13	What requirements would an ideal topoplot need to meet?
Closing	
Q14	Is there anything I did not ask about that you think is important regarding topoplots or uncertainty visualization?

Table S1: Semi-structured interview guide used in nine qualitative interviews.